Machine Learning Canvas

Product:	Authors:	Date:	Version:
VaLoRe	Antoine, Hoang Linh, Duy Vu	March 29, 2026	V1

Background

The main user need is simple: estimating a property's expected price quickly and consistently. In real life, price estimation can take time, may depend too much on manual judgment, and can vary between people.

Value proposition

- For buyers: quickly estimate fair property price
- For sellers: set competitive listing prices
- For real estate agents: support data-driven pricing decisions
- For banks / investors: assist in property risk and valuation analysis

Objectives

- Explore and process the given data
- Build a regression model
- Serve the model through API
- Deploy an end-to-end system on Cloud that allow CI/CD, automated model training, model serving, online prediction, performance monitoring, automated model retraining.

Solution

A MLOps application with two main sides, including serving and training

Feasibility

- Data is tabular with moderate data size
- ML techniques performs well on tabular data and is practical to serve
- Cloud stack is manageable because it stays within one ecosystem, Google Cloud

- Potential risk: quality of the data

Data

Housing Price in King County, USA between May 2014 and May 2015
- Source: Kaggle
- Datapoints: 21,614
- Data type: tabular / structured
- Number of features: 21, including `price`
- Target variable: price
- No missing values and duplicates

Based on the aim of the project, the dataset will be stored in BigQuery as the main cloud data source, and the raw CSV exist in GCS

Metrics

RMSE, MAE, R^2

Evaluation

Offline: LS / TS splits
- Model selection: K-fold CV
- Model evaluation: Hold-out method

Online: Control deployment and feedback

Modeling

Experiment - Baseline model selection
- Data splitting: LS (80%), TS (20%)
- Model zoo: Dummy Regressor, Linear Regression, KNN, Random Forest, XGBoost,etc.
- Hyper-param tuning: K-fold CV
- Model comparison: Hold-out method
  + Train on LS
  + Performance evaluate on TS
  + Baseline model is chosen based on the best performance regarding metric

Model training
- Data splitting: LS (80%), TS (20%)
- Regression model: baseline model
- Model selection: K-fold CV
- Model evaluation: Hold-out method
  + Train on LS
  + Performance evaluate on TS

Inference

Thanks to light-weight ML model, API, container as well as Cloud service, the system could infer the new instance in real-time

Feedback

- Model feedback
- Serving feedback
- Pipeline feedback
- Milestone feedback

Project

Team member:
- Antoine DECKERS
- Hoang Linh BUI
- Duy Vu DINH

- Deliverables: GitHub repo, Dashboard, Report slides, also milestone's objectives

Roadmap:
- Milestone 1:
  + Git, CI/CD, GitFlow setup
  + EDA
  + Preliminary modeling
- Milestone 2:
  + modeling improvement
  + Model serving and deploy for API
  + ML pipeline
- Milestone 3:
  + Dashboard
  + Integration
  + Automation

Machine learning canvas from Made With ML by Goku Mohandas
License: CC BY-SA 4.0